

1. Fundamentals of Quadratic Equations

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Standard Form & Definition

A quadratic equation is a second-degree polynomial equation involving a single variable x . The standard algebraic form is:

$$ax^2 + bx + c = 0 \text{ (where } a \neq 0, \text{ and } a, b, c \text{ are real numbers).}$$

Roots or solutions represent the x -intercepts of the quadratic parabola where $y = 0$.

Core Solution Methods

1. Factorization Method: Used when $ax^2 + bx + c$ can be factored into $(px + q)(rx + s) = 0$.
2. Completing the Square: Moving constant c to RHS, adding $(b/2a)^2$ to both sides.
3. Quadratic Formula: $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$.
4. Graphical Resolution: Plotting parabola and reading off points where curve intersects x -axis.

2. Discriminant & Nature of Roots

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Discriminant Rule: $\Delta = b^2 - 4ac$

- If $\Delta > 0$: Two distinct real roots.
- If $\Delta = 0$: Two coincident/equal real roots.
- If $\Delta < 0$: No real roots (roots are complex conjugate).
- If Δ is a perfect square: Roots are rational numbers.

Worked Example: $2x^2 - 7x + 3 = 0$

Given: $a = 2$, $b = -7$, $c = 3$.

$$\Delta = (-7)^2 - 4(2)(3) = 49 - 24 = 25.$$

$$x = \frac{-(-7) \pm \sqrt{25}}{2 \times 2} = \frac{7 \pm 5}{4}.$$

$$x_1 = \frac{7 + 5}{4} = 3$$

$$x_2 = \frac{7 - 5}{4} = 0.5$$

Final roots: $x = 3$ or $x = 1/2$.

3. Practice Exercises & Exam Strategies

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Practice Questions

1. Solve by factorization: $3x^2 - 10x + 8 = 0$.
2. If α and β are roots of $x^2 - 5x + 6 = 0$, find the value of $\alpha^2 + \beta^2$.
3. Find the value of k for which $x^2 - kx + 9 = 0$ has equal roots.

High-Yield WAEC & JAMB Tips

- Remember that $(\alpha + \beta) = -b/a$ and $\alpha\beta = c/a$.
- Watch your signs! Subtracting a negative b creates a positive numerator.
- When completing the square, ensure the leading coefficient a is normalized to 1 first.